

Responsible Scholarship in Mathematics

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Responsible Scholarship Training in Mathematics

Responsible Scholarship training is required for all PhD students at KU.

- ① Complete the [CITI Responsible Conduct of Research Basic course](#). Send your certificate of completion to Michelle.
- ② One meeting with Jeremy (2/23/26, 4–5pm, Snow 302) to present an overview of topics in responsible scholarship in math (what we are doing now)
- ③ Split into three groups of four students. Each quartet will meet with each of their advisors for one hour to discuss one of three topic sets:
 - Authorship, publication, plagiarism
 - Peer review, refereeing, grant proposal preparation, confidentiality
 - Conflicts of interest, student-mentor relationship
- ④ Each quartet will present a one-hour report to your peers (to be scheduled; often on Stop Day).
- ⑤ Complete the [report form](#) and have your advisor approve it.

“Responsible scholarship is essential for everyone involved in the discovery, application and dissemination of knowledge.”

General resources:

- [Responsible Scholarship at KU](#) (source of above quote)
- [American Mathematical Society \(AMS\) Policy Statement on Ethical Guidelines](#)
- [AMS Policy Statements and Guidelines](#) (See in particular “Information Statements of the AMS Committee on the Profession”)
- [American Statistical Association \(ASA\) Ethical Guidelines for Statistical Practice](#)
- [Prof. Weishi Liu's 2019 RS presentation](#) (reproduced by permission)

- How do you write a research paper? (Well. . .)
- What goes into a paper? (abstract, introduction, definitions, theorems, proofs, bibliography)
- Collaboration: senior vs. junior coauthors, presumption of equality of authors (unlike other fields!), alphabetical rule
- Mathematical writing: [T_EX/L^AT_EX](#), [Knuth–Larrabee–Roberts](#), [Lee](#), [Bertsekas](#), [Su](#), [Tao](#), [Pak](#), . . .
- Literature searches: [MathSciNet](#), [arXiv](#), [Google Scholar](#), [KU Libraries](#), [MathOverflow](#)
- Managing bibliographies: [BibTeX](#), [Zotero](#), . . .

- **A paper is not “official” until it is accepted for publication by a peer-reviewed journal.**
- Before submitting: [arXiv](#), share privately with trusted experts
- Journals vary: general/specialist, selective/less selective, commercial/society, print/electronic. Consult senior colleagues.
- [Open access](#) movement
- Avoid “[predatory](#)” journals. Warning signs: solicitation of submissions by mass email; author charges; not indexed in MathSciNet; overly fast reviews; editorial boards without experts; . . .
- No double submission!
- OK to inquire (politely) about paper status if you do not hear back in ~6 months
- Rejections happen; keep trying!

Plagiarism (DON'T DO IT)

- **cite sources**, consult experts, do not lift large chunks of text without attribution (when in doubt, better to cite than to not cite)
- R. Neidinger, “Avoiding Plagiarism in Mathematics”, Math Horizons, 23 (2016) no. 4, 16-17 [[local copy](#)]
- [Authorial Integrity in Scientific Publication](#) (SIAM)
- [Recommendations for Promoting Integrity in Scientific Journal Publications](#) (Council of Science Editors)
- [“Avoiding Plagiarism, Self-plagiarism, and Other Questionable Writing Practices: A Guide to Ethical Writing”](#) (Office of Research Integrity)
- [StackExchange discussion](#) from 2017
- [A notorious case of math plagiarism](#) (follow the links to primary sources on the Wikipedia page)

All reputable journals, by definition, require manuscripts to undergo peer review before publication.

- *Editor*: receives submissions, decides which merit peer review, finds referee(s), gives guidance to referee about standards to apply (often depends on journal), intermediary for all correspondence, makes final decision to accept or reject
- *Referee*: independent expert responsible for evaluating the merits of a paper (correctness, importance, clarity, appropriate citations), making suggestions for improvement, submitting written recommendation (accepts/revise/reject) to editor; often involved in reviewing subsequent revisions; *does not communicate directly with author*
- All parties (editor, referee, author) should behave professionally: keep the focus on the merits of the manuscript; use neutral and impersonal language when pointing out errors or disagreeing over revisions.

- Typical time frame for review: 4–6 months (varies greatly; much slower in mathematics than most other fields)
- Most math journals use *single-blind* review (referee knows who author is, but not vice versa). Some are *double-blind* (neither referee nor author knows who the other is).
 - Referees must not base their evaluation on the identity of the author(s)
 - Referees must disclose any conflicts of interest to editor
 - Referees must not misuse their knowledge of the results of the paper
 - Referees may count their work as professional service; OK to list journals but not individual papers
- [Best Current Practices for Journals](#) (IMU Committee on Electronic Information and Communication, 2011; published in AMS Notices)
- [How Mathematics Research Journals Select Articles](#), Notices of the AMS **65** (2018), no. 1, 62–64

Grant Proposal Preparation

- *Major funding agencies:* [NSF](#) (though see [this](#)); [NSA](#), [AIM](#), [national labs](#), [Simons](#), [institutes](#) ([SLMath](#), [ICERM](#), [IPAM](#), [Fields](#), ...)
- *Categories of funding:* summer salary; support for graduate students; travel for PI, students, visitors; conference hosting
- Many proposals have two parts:
 - **Scientific merit:** What do you plan to accomplish? What methods will you use? How will it advance mathematical knowledge?
 - **Broader impact:** How will your work benefit society? E.g., solve real-world problems, contribute to education, involve students
- *Preparation/budgeting:* get help from staff, sponsored research office ([KU Office of Research](#))
- *Publication:* Mathematics grants typically encourage you to publish results stemming from the research; always acknowledge sponsor

Conflicts of Interest

“A conflict of interest is a set of circumstances that creates a risk that professional judgement or actions regarding a primary interest will be unduly influenced by a secondary interest.”¹

- **KU policies on COIs**
- **Examples** (many financial in nature) of COIs in academia. Also:
- Other (non-financial) common examples: nepotism, conflict of commitment, inappropriate use of professional position, . . .
- In addition to avoiding COIs, even the appearance of a COI can be damaging. Err on the safe side!
- Prevention:
 - **remove yourself** from situations where you may have a conflict
 - **disclose the conflict** to the person in charge (chair, grant officer, editor, etc.)

¹Lo and Field, *Conflict of Interest in Medical Research, Education, and Practice*, National Academic Press, 2009

Maintenance of Confidentiality

Most online sources about confidentiality in research focus on methods to keep confidential the identity of participants in research studies — typically not an issue in mathematics. However, confidentiality is important in many contexts:

- As author/coauthor (consult coauthors before disseminating research)
- As advisor (guard student research and concerns)
- As student (materials and communications shared by advisor)
- As colleague (respect privacy of personal communications)
- As teacher (student names, grades, IDs, contact info, work)
- As referee/grant reviewer (communicate only with editor/granting agency unless authorized)

Ask before disseminating!

Student-Mentor Relations and Responsibilities

- [KU COGA mentoring guide](#)
- CLAS Mentoring Agreement Template ([PDF](#) | [Word](#))
- [Graduate Student Mentoring Guide: A Guide for Students](#) (U. Michigan Graduate School)

- [Advice to a Young Mathematician](#) by Atiyah, Bollobás, Connes, McDuff, Sarnak
- [Advice for New Doctoral Advisors](#), AMS blog on Teaching and Learning Mathematics
- [The Professor/Grad Relationship](#)

Responsible Use of Artificial Intelligence

- AI and its widespread use is a recent development
- Can be a resource in teaching, research, writing, ...
- Fast-developing area; currently no professional consensus about the boundary between appropriate and inappropriate use

A sample of materials out there:

- [KU CTE: AI Resources](#)
- “The Mathematician’s Assistant: Integrating AI into Research Practice” by J. Henkel, August 2025
- AMS policy on AI use in journals, adopted 2023
- Caltech group “focused on developing AI systems that can tackle hard research-level math problems”
- NY Times article quoting top mathematicians on use of AI as a research tool